

BAKER, MT, USA

WMO: 726777

Lat: 46.346N

Lon: 104.258W

Elev: 2971

StdP: 13.19

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 94055

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.9	(c) -9.1	(d) -20.0	(e) 2.0	(f) -14.2	(g) -14.8	(h) 2.7	(i) -9.1	(j) 32.8	(k) 20.6	(l) 30.3	(m) 22.7	(n) 8.0	(o) 280	(p) 0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.3	95.0	65.2	91.2	64.8	88.1	64.1	70.2	86.4	68.1	84.0	66.2	81.9	11.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	102.5	76.3	63.1	96.7	74.2	61.2	90.1	72.1	36.0	86.5	34.0	83.8	32.5	81.7	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.2	26.5	23.9	-23.9	101.2	4.8	3.6	-27.4	103.8	-30.2	105.9	-32.9	108.0	-36.4	110.6
			-24.3	74.0	4.7	3.7	-27.7	76.6	-30.4	78.8	-33.0	80.9	-36.4	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	19.5	19.9	32.0	42.7	52.4	62.7	71.8	69.2	59.1	44.5	32.7	20.6		
	DBStd	21.16	13.28	13.95	13.20	9.52	8.35	7.12	6.32	6.61	8.97	9.42	11.35	13.46		
	HDD50	4351	946	844	565	249	72	3	0	21	218	522	911			
	HDD65	8139	1411	1264	1023	668	397	125	14	35	220	637	969	1376		
	CDD50	2185	0	0	7	31	146	383	676	597	295	47	3	0		
	CDD65	496	0	0	0	1	5	54	225	167	43	1	0	0		
	CDH74	7406	0	0	3	34	159	818	3065	2409	855	62	1	0		
	CDH80	3340	0	0	0	7	30	274	1513	1138	364	14	0	0		
	WSAvg	10.9	11.1	10.4	11.7	12.5	12.2	11.2	10.2	10.0	10.0	10.8	10.5	10.6		
	PrecAvg	14.4	0.4	0.4	0.6	1.4	2.8	2.8	1.9	1.3	1.3	1.2	0.5	0.3		
	PrecMax	20.6	1.3	1.7	1.9	4.0	7.4	7.4	7.3	4.2	3.1	3.7	3.0	0.9		
(15) Precipitation	PrecMin	8.7	0.0	0.0	0.1	0.3	0.9	0.6	0.2	0.1	0.3	0.3	0.0	0.0		
	PrecStd	2.9	0.3	0.3	0.4	0.9	1.8	1.5	1.4	0.9	0.8	0.9	0.6	0.2		
	DB	51.0	56.8	71.8	78.9	83.9	92.6	100.7	99.2	93.5	81.1	68.4	54.0			
	MCWB	40.1	43.8	50.2	53.6	60.6	65.3	67.6	66.7	63.2	57.3	49.5	42.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	44.2	47.7	63.5	70.8	78.6	87.4	96.3	94.2	89.5	73.0	60.8	46.3			
	2% MCWB	36.8	39.1	46.9	51.5	59.3	65.3	65.9	64.7	61.8	53.1	45.7	37.6			
	5% DB	40.1	42.4	57.0	64.8	74.0	82.3	92.5	90.4	84.0	66.2	54.6	42.0			
	5% MCWB	34.5	35.8	43.4	48.8	57.4	63.8	65.6	64.5	59.9	50.6	42.8	34.9			
	10% DB	37.0	37.5	50.9	60.8	69.9	78.7	88.7	86.9	78.5	61.2	49.3	37.4			
	10% MCWB	32.5	32.9	41.1	46.7	55.1	61.6	65.1	63.8	58.1	48.3	39.9	32.5			
	0.4% WB	41.0	44.8	51.1	56.0	65.3	71.4	74.1	72.7	66.7	59.0	50.3	42.6			
	0.4% MCDB	49.5	55.8	69.7	75.0	77.0	83.7	88.5	89.1	84.5	77.3	66.3	52.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	37.3	39.3	47.6	52.5	61.7	68.3	70.9	69.1	64.2	55.1	46.5	38.2			
	2% MCDB	43.1	47.0	62.2	67.9	74.2	81.3	87.5	86.9	81.6	68.8	59.8	45.4			
	5% WB	35.0	36.0	44.4	49.9	58.9	65.8	68.7	67.1	62.2	51.9	43.1	35.5			
	5% MCDB	39.8	41.6	56.1	63.3	71.0	78.4	84.9	83.7	79.0	64.8	53.6	41.6			
	10% WB	32.9	33.4	41.3	47.6	56.3	63.6	66.7	65.3	59.4	49.1	40.2	33.0			
	10% MCDB	36.9	37.5	50.3	59.1	67.7	75.6	82.8	81.0	75.3	60.0	48.5	37.4			
	MDBR	19.8	19.9	21.9	23.4	23.3	24.1	28.3	28.8	27.6	23.7	22.3	19.5			
	5% DB	21.4	24.1	32.2	33.0	31.1	31.0	34.3	35.1	37.5	34.0	31.6	24.2			
(36) Mean Daily Temperature Range	5% WB	15.7	16.7	18.0	17.3	15.5	14.0	12.1	13.2	14.9	17.7	18.8	16.4			
	MCDBR	20.8	23.1	31.1	30.3	27.5	27.5	29.9	29.4	32.4	31.3	30.3	23.1			
	MCWBR	15.7	16.6	17.8	16.6	14.6	13.8	12.6	13.1	13.8	16.7	18.3	16.2			
	taub	0.250	0.256	0.273	0.307	0.328	0.341	0.353	0.355	0.320	0.288	0.259	0.246			
(40) Clear-Sky Solar Irradiance	taud	2.546	2.556	2.562	2.458	2.462	2.458	2.449	2.416	2.500	2.572	2.547	2.544			
	Ebn at Noon	280	300	308	302	297	292	287	283	285	282	273	269			
	Edh at Noon	19	23	27	33	34	34	34	34	29	23	19	17			
	RadAvg	531	836	1207	1574	1859	2141	2266	1878	1414	906	580	421			
	RadStd	42	60	92	84	182	147	84	108	101	79	61	23			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page